A Seniority-Resolved Nonlinear Canonical Transformation for Spin-$\frac{1}{2}$ Fermions

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We introduce a seniority-resolved nonlinear canonical transformation of spin-$\frac{1}{2}$ fermions (electrons). The central motivation for this work is the observation that seniority—the number of unpaired electrons—provides a natural organizing principle for correlated electronic systems. By interpreting seniority as an operator grading of the local Clifford algebra, we partition the on-site $su(4)$ algebra written in terms of Majoranas into seniority-preserving and seniority-changing sectors. We then selectively apply the Jordan–Wigner (JW) transformation to the seniority-preserving $su(2)$ subalgebra mixing spin and charge, resulting in a nonlinear but globally canonical automorphism of the fermionic operator algebra without constraints. Mapped one-body operators become many-body operators due to JW strings, yet their matrix elements can still be evaluated at polynomial cost. Proof-of-principle benchmarks on the repulsive Hubbard model demonstrate that this Majorana–Jordan–Wigner (MJW) transformation yields mean-field states with large overlap with the exact ground state and that the transformed Hamiltonian is amenable to weakly correlated treatments across interaction strengths.

I. INTRODUCTION

The concept of an electron pair permeates the entirety of chemistry and a large portion of condensed matter physics. From bonding [Pauling book] to superconductivity [BCS paper], electron pairs play a crucial role in the description of physical phenomena. While electrons obey fermionic algebra, electron pair operators obey the $su(2)$ Lie algebra and behave as hard-core bosons.

The Jordan–Wigner (JW) transformation [1] allows one to convert Hamiltonians built from $su(2)$ operators into fermionic Hamiltonians, and vice versa. In recent work [2–5], we have shown that when mapping with JW from spins to fermions, one increases the mean-field overlap with the exact ground state, indicating a reorganization of correlations into perturbative form. What we mean by this is that the mean-field solution of the fermionic JW-transformed Hamiltonian has a significantly larger overlap with the exact solution and is energetically more accurate than the mean-field solution of the original spin Hamiltonian. Similar results are obtained for model Hamiltonians based on $su(2)$ pairing generators (cite RBCS). A larger overlap of the mean-field state with the exact ground state implies a smaller degree of entanglement, thus facilitating its treatment with perturbative methods.

The main objective of this work is to extend these ideas to electronic Hamiltonians, composed of spin-$\frac{1}{2}$ fermionic operators and containing not one but two intertwined $su(2)$ algebras: a pairing (charge) channel associated with doubly occupied and empty orbitals, and a spin channel associated with singly occupied orbitals. These two algebras close to $so(4)$ and naturally mix spin and charge degrees of freedom, a structure well known in the context of Hubbard physics and η-pairing symmetry [6].

Exact local mappings of spin-$\frac{1}{2}$ fermions into a tensor product of a spinless fermion and a two-level system have appeared previously in several contexts, including constraint-free spin–charge representations and canonical fermionic reformulations [7–10]. However, these constructions were not framed in terms of seniority nor organized according to a seniority-resolved operator grading.

In this work, we introduce a *nonlinear* canonical transformation between spin-$\frac{1}{2}$ fermions that is naturally organized by seniority. Seniority, defined here as the number of unpaired electrons (broken electron pairs), plays a central structural role.

While exact local mappings of spin-$\frac{1}{2}$ fermions into a tensor product of a spinless fermion and a two-level system have appeared previously, here we reinterpret this structure through the lens of seniority. In our construction, the occupation of an auxiliary fermion directly counts broken pairs, thereby endowing the local Clifford algebra with a natural seniority grading. This grading partitions the on-site $su(4)$ algebra into seniority-preserving and seniority-changing sectors. We then apply Jordan–Wigner fermionization selectively to the seniority-preserving subalgebra, resulting in a nonlinear but globally canonical automorphism of the fermionic operator algebra.

When the underlying Hamiltonian describes a purely spin or purely pairing $su(2)$ model, our transformation reduces to the standard JW form. For general electronic Hamiltonians that break seniority, however, the transformation mixes spin and pairing channels in a controlled algebraic manner.

The transformation we construct—termed the Majorana–Jordan–Wigner (MJW) mapping—is unitary and free of constraints. It can be viewed as a nonlinear automorphism of the fermionic Clifford algebra, implemented globally through Jordan–Wigner strings. Mapped one-body operators become many-body operators due to the strings, yet their matrix elements can still be evaluated at polynomial cost, as shown in our previous work [cite JWHF].

Although our primary emphasis is algebraic, the construction has practical implications. In proof-of-principle calculations for the repulsive Hubbard model [11], the MJW-transformed Hamiltonian yields a mean-field state with overlap comparable to that of doubly-occupied configuration interaction (DOCI) [12–18], while remaining amenable to weakly correlated treatments.

The perspective developed here emphasizes seniority as the structural basis for the transformation rather than as a post hoc truncation criterion. By resolving the local Clifford algebra according to seniority and selectively fermionizing the seniority-preserving subalgebra, we obtain a globally canonical nonlinear transformation that intertwines spin and pairing channels in a natural way. This construction differs from previous spin–charge decompositions in that seniority is not merely an auxiliary label but the organizing principle of the operator algebra itself. Previous mappings did not organize the Clifford algebra according to seniority nor construct a global nonlinear automorphism with selective JW fermionization. While the transformation reduces to the standard Jordan–Wigner mapping in purely spin or purely pairing limits, it provides a unified algebraic framework in which spin and pairing symmetries are treated on equal footing in general electronic Hamiltonians. The novelty of the present work lies not in the existence of a local spin–charge mapping, but in the construction of a global nonlinear canonical automorphism organized by seniority grading, implemented through selective Jordan–Wigner fermionization.

II. MOTIVATION

We begin with an extended discussion that explains the underlying thought process and the previous work that led us to the framework presented in this paper.

Seniority, defined as the number of unpaired electrons, is a useful measure for analyzing electron pairing. In a state with seniority zero, all electrons are paired; a state with one broken pair has seniority two, and so on.

In the present work, it is useful to introduce an orbital-resolved indicator of broken pairing. In our mapping, the occupation $b_p^\dagger b_p \in \{0, 1\}$ flags whether the spatial orbital p is singly occupied (and thus contributes to broken pairing). Because breaking one pair produces two unpaired electrons, the conventional seniority satisfies the even-parity relation $\Omega = 2\sum_p b_p^\dagger b_p$. The pairing $su(2)$ channel, in this framework, has seniority zero, while the spin $su(2)$ channel instead has seniority N, where N is the number of electrons. Hamiltonians built with purely spin and/or pairing operators (vide infra) do not change seniority.

It is important to note that seniority is an orbital-dependent property because orbital rotations change the pairing scheme and, therefore, the seniority number. Methods based on seniority rely on orbital selection to define the pairing scheme, and if the orbitals are rotated, the seniority numbers change. Orbital optimization is crucial for obtaining high-quality results in practice.

Electronic systems possess two distinct seniority-preserving channels: spin and pairing, depending on whether the coupling between electrons occurs along particle-hole (spin) or particle-particle (pairing) channels [cite pECCD]. These two $su(2)$ algebras close to $so(4)$ and act within fixed-seniority sectors. For two-body Hamiltonians, in addition to these seniority-preserving onsite interactions, there are hopping terms that preserve global but not local seniority, and terms that change seniority by two or four units [cite CCD0].

Throughout this work, we will refer to operators that commute with the seniority operator (i.e., that do not change Ω) as seniority-preserving. This terminology should not be confused with the specific sector $\Omega = 0$; rather, it denotes invariance under the seniority grading introduced above.

The coexistence of two independent local seniority-preserving $su(2)$ algebras—spin and pairing—suggests a deeper structural organization of the local fermionic operator algebra. Since these two $su(2)$ sectors close to $so(4)$, the full local $su(4)$ algebra can be partitioned into generators that preserve seniority and generators that change it. This observation provides the algebraic foundation for the canonical transformation introduced later: rather than treating seniority as a truncation criterion for wave functions, we use it to reorganize the operator algebra itself.

In this sense, seniority defines a $\mathbb{Z}_2$-graded structure on the local operator algebra, distinguishing generators that preserve broken-pair parity from those that change it.

In previous work [18, 19], we have shown that strongly correlated molecular dissociations are well described by low-seniority wave functions, especially when using orbitals that break point-group symmetry. Low seniority contains high-order excitations with a low number of broken pairs, highlighting the importance of the seniority hierarchy, rather than excitation degree, in describing strong correlations. In particular, the zero-seniority DOCI wave function, obtained by exact diagonalization of the seniority-conserving part of the Hamiltonian with a suitably chosen pairing scheme, is qualitatively correct along entire dissociation curves for molecules such as N_2 and in the atomization of hydrogen clusters. The addition of determinants from low-seniority sectors rapidly converges to the exact answer.

Unfortunately, DOCI is combinatorial and therefore

intractable for large systems. A major step forward in bypassing this combinatorial barrier resulted from the realization that for electronic systems, orbital-optimized DOCI is very well approximated by orbital-optimized pair-Coupled-Cluster-Doubles (pCCD) [? ?], a wave function that can be obtained with low polynomial cost. pCCD captures the essential physics of strong static correlation in many molecular dissociations by incorporating high-order excitations while restricting to configurations that preserve seniority. Moreover, pCCD provides a natural reference for systematically improving accuracy by incorporating additional correlation effects, either through perturbative corrections or by coupling to higher-seniority sectors in a controlled way [cite fpCCD / pCCD extensions as appropriate]. At the same time, experience with pCCD and related seniority-based approaches highlights an important limitation: when the reference state is qualitatively mismatched to the target state, the remaining correlation can become strongly nonperturbative and convergence can deteriorate [cite one or two representative examples from our work]. This motivates the central question of the present work: rather than approximating the seniority structure within the original fermionic variables, can one construct a canonical transformation that reorganizes the operator algebra itself according to seniority?

These observations motivate a different question: rather than approximating the seniority-zero sector within the original fermionic representation, can one construct a canonical transformation that reorganizes the full fermionic algebra according to seniority? If such a transformation exists, it would provide an alternative route to weakly correlated descriptions by embedding the seniority hierarchy directly into the operator structure. The Majorana–Jordan–Wigner construction introduced below is the realization of this idea.

III. THEORY

A. Preliminaries

1. *Seniority Zero Channels of Electronic Hamiltonians*

We define the seniority operator as the number of unpaired electrons, i.e., the number of electrons minus twice the number of doubly occupied orbitals,

$$\Omega = \sum_p \left(N_p - 2\,D_p\right), \tag{1}$$

where D_p is 1 if level p is doubly occupied and 0 otherwise. In terms of spin-$\frac{1}{2}$ physical fermions, we have

$$P_p^\dagger = c_{p\uparrow}^\dagger c_{p\downarrow}^\dagger, \quad n_{p\uparrow} = c_{p\uparrow}^\dagger c_{p\uparrow}, \quad n_{p\downarrow} = c_{p\downarrow}^\dagger c_{p\downarrow}. \tag{2}$$

$$D_p = P_p^\dagger\, P_p = n_{p\uparrow}\, n_{p\downarrow} \tag{3}$$

$$N_p = n_{p\uparrow} + n_{p\downarrow} \tag{4}$$

$$\Omega = \sum_p \left(n_{p\uparrow} - n_{p\downarrow}\right)^2, \tag{5}$$

after using $n^2 = n$ for fermions. Ω is a quadratic operator in the original fermions but becomes linear in the auxiliary fermions introduced below. For the local states $|-\rangle, |\uparrow\rangle, |\downarrow\rangle, |\uparrow\downarrow\rangle$, the corresponding contributions to Ω are $0, 1, 1, 0$, respectively; thus Ω counts broken pairing (singly occupied orbitals) in the chosen one-particle basis. Fully paired configurations have $\Omega = 0$, whereas configurations with one unpaired electron on every site have $\Omega = M$ for M orbitals. Because the pairing scheme depends on the orbital basis, orbital rotations generally change Ω and can mix seniority sectors. Seniority-based truncations and diagnostics have therefore been particularly useful in quantum chemistry [18].

Electronic Hamiltonians admit two canonical seniority-preserving $su(2)$ subalgebras: spin and pairing. They are defined by

$$S_p^+ = c_{p\uparrow}^\dagger\, c_{p\downarrow}, \quad S_p^- = c_{p\downarrow}^\dagger\, c_{p\uparrow}, \quad S_p^Z = \frac{1}{2}\left(n_{p\uparrow} - n_{p\downarrow}\right) \tag{6}$$

and

$$P_p^\dagger = c_{p\uparrow}^\dagger\, c_{p\downarrow}^\dagger, \quad P_p = c_{p\downarrow}\, c_{p\uparrow}, \quad N_p = n_{p\uparrow} + n_{p\downarrow}, \tag{7}$$

respectively. They each satisfy the $su(2)$ commutation relations modulo the shifting and scaling of the Cartan generator, i.e.,

$$[S_p^+, S_q^-] = 2\,\delta_{pq}\, S_p^Z, \quad [S_p^Z, S_q^\pm] = \pm\delta_{pq}\, S_p^\pm, \tag{8}$$

and

$$[P_p^\dagger, P_q] = \delta_{pq}\,\left(N_p - 1\right), \quad [N_p, P_q^\dagger] = 2\,\delta_{pq}\, P_p^\dagger. \tag{9}$$

These two $su(2)$ representations commute with each other and with the seniority operator Ω, and therefore act within fixed-seniority sectors. In contrast, generic one-body hopping and exchange operators (e.g., $c_{p\uparrow}^\dagger c_{q\uparrow}$ for $p \neq q$) generally change seniority and do not commute with Ω. We want to emphasize here the difference between local versus global seniority conserving operators; MJW is local, a bijection of the operator algebra; the $so(4)$ component defines operators that preserve local seniority, while the rest of the fermionic algebra contains operators that preserve global but not local seniority.

2. *Majorana Operators and Clifford Algebras*

To expose the underlying algebraic structure, we now switch from creation/annihilation operators to Majorana generators. We define the Majoranas

$$\gamma_{1p} = c_{p\uparrow} + c_{p\uparrow}^\dagger, \qquad \gamma_{2p} = \mathrm{i}\left(c_{p\uparrow} - c_{p\uparrow}^\dagger\right),$$
$$\gamma_{3p} = c_{p\downarrow} + c_{p\downarrow}^\dagger, \qquad \gamma_{4p} = \mathrm{i}\left(c_{p\downarrow} - c_{p\downarrow}^\dagger\right), \tag{10}$$

which satisfy

$$\{\gamma_{\mu p}, \gamma_{\nu q}\} = 2\,\delta_{pq}\,\delta_{\mu\nu} \quad \mu,\nu = 1-4, \quad p,q = 1-M,$$
$$\gamma_{\mu p}^\dagger = \gamma_{\mu p}, \qquad \gamma_{\mu p}^2 = 1. \tag{11}$$

The set $\{\gamma_{\mu p}\}$ generates a Clifford algebra: the linear span of all Majorana monomials (products of distinct Majoranas), together with the identity. We refer to these monomials as Clifford elements. Their multiplication rules are fixed by the canonical anticommutation relations above. For a single site p, Cl(4) has dimension $2^4 = 16$; excluding the identity, the remaining 15 Hermitian elements provide a basis for the local $su(4)$ algebra. For a given site, we have

$$\begin{aligned}
&\gamma_1,\ \gamma_2,\ \gamma_3,\ \gamma_4,\\
&\gamma_1\,\gamma_2,\ \gamma_1\,\gamma_3,\ \gamma_1\,\gamma_4,\ \gamma_2\,\gamma_3,\ \gamma_2\,\gamma_4,\ \gamma_3\,\gamma_4,\\
&\gamma_1\,\gamma_2\,\gamma_3,\ \gamma_1\,\gamma_2\,\gamma_4,\ \gamma_1\,\gamma_3\,\gamma_4,\ \gamma_2\,\gamma_3\,\gamma_4,\\
&\gamma_1\,\gamma_2\,\gamma_3\,\gamma_4.
\end{aligned}$$

The commutator algebra of these generators reproduces the structure constants of $su(4)$. The usual fermion operators can be obtained from Majoranas as:

$$c_{p\uparrow}^\dagger = \frac{1}{2}\,(\gamma_{1p} + \mathrm{i}\,\gamma_{2p}), \tag{12a}$$
$$c_{p\downarrow}^\dagger = \frac{1}{2}\,(\gamma_{3p} + \mathrm{i}\,\gamma_{4p}). \tag{12b}$$

With these preliminaries out of the way, we can proceed with the definition of our Majorana-Jordan-Wigner transformation.

B. The First Step: Seniority Mapping

We begin by partitioning the all-orbital Clifford algebra into two sets that preserve and change seniority, respectively, and generate the entire Fock space. Recall that the six onsite bilinear Majorana products, $\gamma_{\mu p}\gamma_{\nu p}$, close $so(4)$

$$su(2)_{spin} + su(2)_{pairing} = so(4) \tag{13}$$

This subalgebra is maximal among local seniority-preserving operators. We next select *any* three Majoranas, e.g., $\{\gamma_{1p}, \gamma_{2p}, \gamma_{3p}\}$, and use their bilinear products to construct three Pauli operators that close $su(2)$:

$$\sigma_p^X = \mathrm{i}\,\gamma_{2p}\,\gamma_{3p}, \quad \sigma_p^Y = \mathrm{i}\,\gamma_{3p}\,\gamma_{1p}, \quad \sigma_p^Z = -\mathrm{i}\,\gamma_{1p}\,\gamma_{2p},$$
$$\left[\sigma_p^a, \sigma_p^b\right] = 2\,\mathrm{i}\,\varepsilon_{abc}\,\sigma_p^c. \tag{14}$$

We will hereafter refer to the σ operators as Paulis. By selecting one of the four distinct trios of Majoranas, we can construct four possible $su(2)$ algebras. All of them irremediably mix *physical spin* and *physical pairing* generators. For the case under consideration, the Paulis are

$$\begin{aligned}
\sigma_p^X &= S_p^+ + S_p^- + P_p^\dagger + P_p,\\
\mathrm{i}\,\sigma_p^Y &= S_p^+ - S_p^- + P_p^\dagger - P_p,\\
\sigma_p^Z &= 2\,S_p^Z + N_p - 1,
\end{aligned}$$

as can be seen by substituting the Paulis with the products of Majoranas in Eqn. (14), using the definitions of the Majorana operators provided in Eqn. (10), and then recognizing the zero-seniority operators of Eqns. (6)–(2). These Paulis are invariant to Z_2 conjugation of the down fermion, $c_{p\downarrow}^\dagger \to c_{p\downarrow}$, a unitary transformation that interchanges spin with pairing [cite Nambu].

The next step utilizes the product of three Majoranas into a so-called emergent Majorana

$$\gamma_{0p} = i\gamma_{1p}\gamma_{2p}\gamma_{4p}, \quad \gamma_{0p}^\dagger = \gamma_{0p}, \quad \gamma_{0p}^2 = 1$$

and combines it with γ_{3p}, to build a spinless fermion:

$$\begin{aligned}
b_p^\dagger &= \frac{1}{2}\,(\gamma_{3p} + \mathrm{i}\,\gamma_{0p}), \qquad & b_p &= \frac{1}{2}\,(\gamma_{3p} - \mathrm{i}\,\gamma_{0p}),\\
\gamma_{3p} &= b_p + b_p^\dagger, & \gamma_{0p} &= \mathrm{i}\,\left(b_p - b_p^\dagger\right).
\end{aligned} \tag{15}$$

We can then write

$$\begin{pmatrix} c_{p\uparrow}^\dagger \\ c_{p\downarrow}^\dagger \end{pmatrix} = \frac{1}{2}\begin{pmatrix} \sigma_p^+\,\gamma_{3p} \\ \gamma_{3p} - \mathrm{i}\,\gamma_{0p}\,\sigma_p^z \end{pmatrix} \tag{16a}$$
$$= \frac{1}{2}\begin{pmatrix} \sigma_p^+\,\left(b_p + b_p^\dagger\right) \\ \left(b_p + b_p^\dagger\right) + \left(b_p - b_p^\dagger\right)\,\sigma_p^Z \end{pmatrix} \tag{16b}$$

where $\sigma_p^+ = \sigma_p^X + \mathrm{i}\,\sigma_p^Y$. The mapping above can be verified by direct replacement. The physical electronic states become the direct product of a set of Paulis times a set of spinless fermions:

$$\begin{aligned}
|-\rangle &\mapsto |\Downarrow\rangle \otimes |0\rangle,\\
|\downarrow\rangle &\mapsto |\Downarrow\rangle \otimes |1\rangle,\\
|\uparrow\rangle &\mapsto |\Uparrow\rangle \otimes |1\rangle,\\
|\uparrow\downarrow\rangle &\mapsto |\Uparrow\rangle \otimes |0\rangle.
\end{aligned}$$

Here, $|\Downarrow\rangle$ and $|\Uparrow\rangle = \frac{1}{2}\sigma^+\,|\Downarrow\rangle$ are $su(2)$ states, and $|0\rangle$ and $|1\rangle = b^\dagger\,|0\rangle$ are fermionic states. Using the Majorana representation, it is simple to check that the Paulis and Majoranas commute at different sites, but anticommute at the same site. This first step, eqn (13), is a *seniority* mapping because the $su(2)$ preserves physical seniority, whereas $b_p^\dagger$ increases it by 1, i.e., the seniority of an electronic state is equal to the number of b fermions present in the mapped representation. Previous spin–charge mappings [Chandler & Gunn, Kumar, Bazzanella] constructed commuting spinless fermions and su(2) generators. In contrast, our construction enforces onsite fermionic anticommutation between the su(2) sector and the spinless fermion, a structural requirement for the subsequent Jordan–Wigner step to yield a globally canonical transformation.

C. The Second Step: Jordan-Wigner Mapping

The second step in MJW maps the Paulis to a second set of spinless fermions via the JW transformation.

The significance of this step is that mean-field theory in the mapped representation becomes equivalent to a correlated theory in the physical seniority zero frame [2]. This second step builds on the correlation-weakening that JW provides for the seniority-zero channel [2–4], which is the main motivation for our construction. Treating the zero seniority channel with JW is expected to alleviate problems previously identified as the main source of trouble in the accurate treatment of strongly correlated electronic systems [18, 19].[cite Jacob DOCI vs FCI0]

The Jordan-Wigner transformation maps Paulis to a set of spinless fermions with creation and annihilation operators $(a^\dagger, a)$ as

$$\begin{aligned} \sigma_p^+ &\mapsto 2\, a_p^\dagger\, \phi_p^a, \\ \sigma_p^- &\mapsto 2\, a_p\, \phi_p^a, \\ \sigma_p^Z &\mapsto 2\, a_p^\dagger\, a_p - 1, \end{aligned} \tag{17}$$

where the string operator ϕ_p^a is written in terms of a fermions

$$\phi_p^a = \exp\left(\mathrm{i}\,\pi \sum_{k<p} a_k^\dagger\, a_k\right). \tag{18}$$

We can define a second set of Majoranas, denoted $\eta_{\mu p}$ and defined in analogy to $\gamma_{\mu p}$ but in terms of the a fermions rather than the physical ones:

$$\eta_{1p} = a_p + a_p^\dagger, \qquad \eta_{2p} = \mathrm{i}\,\left(a_p - a_p^\dagger\right). \tag{19}$$

In terms of these Majorana operators, we have

$$\sigma_p^+ \mapsto \left(\eta_{1p} + \mathrm{i}\,\eta_{2p}\right)\, \phi_p^a, \tag{20}$$

$$\sigma_p^- \mapsto \left(\eta_{1p} - \mathrm{i}\,\eta_{2p}\right)\, \phi_p^a, \tag{21}$$

$$\sigma_p^Z \mapsto 1 - \mathrm{i}\,\eta_{1p}\,\eta_{2p}, \tag{22}$$

$$\phi_p^a = \prod_{k<p} \left(\mathrm{i}\,\eta_{1k}\,\eta_{2k}\right). \tag{23}$$

Because a fermions anticommute with b fermions, an additional string operator χ_p is needed to guarantee that the mapped Paulis continue commuting with the b fermions:

$$\chi_p = \exp\left(\mathrm{i}\,\pi \sum_{\substack{1\le k\le M,\\ k\neq p}} b_k^\dagger\, b_k\right) = \Pi_b\, \left(1 - 2\, b_p^\dagger\, b_p\right), \tag{24}$$

where $\Pi^b = \exp\left(\mathrm{i}\,\pi \sum_{k=1}^M b_k^\dagger\, b_k\right)$ is the global number-parity operator of b fermions, known as a Klein factor, whose associated Majoranas are

$$\eta_{3p} = b_p + b_p^\dagger, \qquad \eta_{4p} = \mathrm{i}\,\left(b_p - b_p^\dagger\right). \tag{25}$$

Finally, the correspondence between physical and dual fermion states becomes

$$|-\rangle \mapsto |-\rangle\,, \tag{26}$$

$$|\downarrow\rangle \mapsto |b\rangle\,, \tag{27}$$

$$|\uparrow\rangle \mapsto |ab\rangle\,, \tag{28}$$

$$|\uparrow\downarrow\rangle \mapsto |a\rangle\,. \tag{29}$$

This dictionary of states between the two representations fixes the gauge of the MJW transformation.

D. MJW Mapping

After incorporating Π^b into the mapping, we finally write

$$\begin{pmatrix} c_{p\uparrow}^\dagger \\ c_{p\downarrow}^\dagger \end{pmatrix} \mapsto \frac{1}{2} \begin{pmatrix} 2a_p^\dagger\left(b_p^\dagger - b_p\right)\phi_p^a \Pi^b \\ \left(b_p + b_p^\dagger\right) - \left(b_p - b_p^\dagger\right) e^{\mathrm{i}\pi a_p^\dagger a_p} \end{pmatrix} \tag{30}$$

Canonical anticommutation relations (CAR) readily follow using the expressions on the right-hand side above. Therefore, the MJW mapping defines an automorphism of the fermionic CAR algebra.

We note that Π^b plays no role in the theory unless we carry out the MJW mapping in a basis that breaks global S_Z symmetry, i.e., a noncollinear basis where not all fermionic terms have an even number of b fermions. Mapping in a noncolinear basis with broken S_Z is legitimate, but will additionally require $so(2M+1)$ mean-field tools, as discussed in previous work [cite]. In this paper, we will restrict ourselves to Hamiltonians written in a collinear orbital basis, so we will not carry Π_b forward in the derivations below. Nevertheless, the formalism presented hereafter remains valid on a noncollinear basis with some minimal adjustments.

Written in this way, MJW is a nonlinear *canonical* transformation between electrons. The nonlinearity arises because linear Majorana generators are mapped to cubic Clifford elements. It is also nonlocal and all-body because of JW strings ϕ_p^a. However, if we ignore strings, which we can treat efficiently with polynomial cost tools, the up fermions increase rank (i.e., polynomial degree) by half-body, whereas down fermions preserve rank. Again, modulo strings, two-body physical interactions in the down-channel remain two-body, but those involving two up electrons increase rank to 3-body, and those between all up electrons become 4-body. The density matrices factorize along a and b channels, which is computationally convenient.

E. MJW as a Clifford Algebra Transformation

Written in terms of Majoranas, MJW is

$$\begin{pmatrix} \gamma_{1p} \\ \gamma_{2p} \\ \gamma_{3p} \\ \gamma_{4p} \end{pmatrix} \mapsto \begin{pmatrix} i\eta_{2p}\eta_{3p}\phi_p^a \\ i\eta_{3p}\eta_{1p}\phi_p^a \\ \eta_{3p} \\ i\eta_{1p}\eta_{2p}\eta_{4p} \end{pmatrix} \tag{31}$$

MJW is a mapping between CA elements of different rank, additionally incorporating JW strings to patch the algebra anticommutation rules between generators $\eta_{\mu,p}$ and $\eta_{\nu,q}$ at different orbitals. The transformed operators are Hermitian, square to the identity, and mutually anticommute, and therefore constitute a new generating set of the same Clifford algebra. Hence, MJW is an automorphism of $Cl(4M)$. For $p = q$, strings play no role, and if we ignore them, MJW is just a rotation in the local $su(4)$, a local automorphism, but a nonlocal nonlinear transformation over the lattice.

A second MJW mapping of the tilde Majoranas recovers the original Majoranas, making it straightforward to transform dual operators into the physical basis. We can additionally write

$$\begin{pmatrix} i\gamma_{2p}\gamma_{3p} \\ i\gamma_{3p}\gamma_{1p} \\ \gamma_{3p} \\ i\gamma_{1p}\gamma_{2p}\gamma_{4p} \end{pmatrix} \mapsto \begin{pmatrix} \eta_{1p}\phi_p^a \\ \eta_{2p}\phi_p \\ \eta_{3p} \\ \eta_{4p} \end{pmatrix} \tag{32}$$

The top two rows above are the JW transformation of the ladder $su(2)$ generators mixing the physical spin and pairing algebras into a fermions, while the bottom two rows describe the transformation of the down physical channel into b fermions. MJW is a natural extension of JW from spinless fermions to up/down physical electrons. Our original motivation for the MJW mapping was to generalize the JW transformation from spinless to Spin-$\frac{1}{2}$ fermions with the aim of weakening strong correlations in the seniority-preserving $su(2)$ algebra (cite) of physical electronic Hamiltonians. Note also that there are three fixed, covariant elements under MJW

$$\begin{aligned} \gamma_3 &\mapsto \eta_3 \\ \gamma_1\gamma_2 &\mapsto \eta_1\eta_2 \\ \gamma_1\gamma_2\gamma_3 &\mapsto \eta_1\eta_2\eta_3 \end{aligned} \tag{33}$$

In summary, the canonical transformation proceeds in three logically distinct steps: (i) identification of a seniority-preserving $so(4)$ subalgebra within the local Clifford algebra; (ii) nonlinear redefinition of Majorana generators consistent with this grading; and (iii) incorporation of Jordan–Wigner strings to maintain global fermionic statistics. Each step preserves the algebraic structure, and together they yield a globally well-defined nonlinear automorphism of the fermionic operator algebra.

Clearly, MJW defines a dual representation that is unitarily equivalent to the physical representation and preserves the Hamiltonian spectrum under exact diagonalization. However, the mean-field approximations differ between the two frames, as discussed below.

F. Seniority Preserving Physical Operators

Let's next examine how physical spin and pairing operators transform under MJW:

$$\begin{aligned} n_{p\uparrow} &\mapsto a_p^\dagger a_p \\ n_{p\downarrow} &\mapsto \left(a_p^\dagger a_p - b_p^\dagger b_p\right)^2 \\ n_{p\uparrow}n_{p\downarrow} &\mapsto a_p^\dagger a_p \left(1 - b_p^\dagger b_p\right) \\ P_p^\dagger &\mapsto a_p^\dagger \left(1 - b_p^\dagger b_p\right) \phi_p^a \\ S_p^+ &\mapsto a_p^\dagger b_p^\dagger b_p \phi_p^a \end{aligned} \tag{34}$$

The mapping for physical particle number is

$$N_p \mapsto 2a_p^\dagger a_p - 2a_p^\dagger a_p b_p^\dagger b_p + b_p^\dagger b_p \tag{35}$$

and for the Z component of spin is

$$S_p^Z \mapsto a_p^\dagger a_p b_p^\dagger b_p - \frac{1}{2} b_p^\dagger b_p \tag{36}$$

Note that the up occupancy $n_{p\uparrow}$ is covariant, but $n_{p\downarrow}$ is not: it becomes the seniority operator, eqn. (1), in the dual representation. Conversely,

$$b_p^\dagger b_p \mapsto (n_{p\uparrow} - n_{p\downarrow})^2 \tag{37}$$

The number of b fermions determines the seniority of the physical state. Equations xx and yy above show that, in a mean-field sense, $\langle b_p^\dagger b_p \rangle$ determines the degree of mixing between the physical spin and pairing algebras. The limit $\langle b_p^\dagger b_p \rangle \to 0$ corresponds to a paired system with negligible spin fluctuations. Vice versa, $\langle b_p^\dagger b_p \rangle \to 1$ corresponds to a spin system where pairing fluctuations are absent. Note that physical number: $N = \sum_p N_p$, and spin: $S^Z = \sum_p S_p^Z$ symmetries are explicitly broken by the transformation. These symmetry operators become two-body, which makes their projection not amenable to standard tools (cite PQT, PHF). On the other hand, a combination of them is conserved:

$$N_\uparrow = \frac{1}{2}N + S^Z \mapsto N_a = \sum_p a_p^\dagger a_p \tag{38}$$

and can be symmetry-projected. Although physical N and S^Z become two-body, physical number and spin parity remain one-body

$$\exp(\ \mathrm{i}\ \pi \sum_p N_p) \mapsto \exp(\ \mathrm{i}\ \pi \sum_p b_p^\dagger b_p) \tag{39}$$

$$\exp(\ \mathrm{i}\ \pi \sum_p S_p^Z) \mapsto \exp(\ \mathrm{i}\ \pi \sum_p b_p^\dagger b_p) \tag{40}$$

after using eqn xx, and can also be symmetry projected.

In Coulombic repulsive systems, electron number N cannot break spontaneously in mean-field theory [cite Lieb]. In the dual representation, N_b is explicitly broken, so both physical number and physical spin-Z are also explicitly broken. However, it remains to be seen if physical Number symmetry can break spontaneously in the dual frame. As already mentioned, $N_a \mapsto N_\uparrow$ is a conserved operator through the transformation.

A last thing to note is that physical seniority-preserving terms like $P_p^\dagger P_q$, when mapped to dual fermions, yield an increase in rank

$$P_p^\dagger P_q \mapsto a_p^\dagger a_q (1 - b_p^\dagger b_p)(1 - b_q^\dagger b_q)\phi_p \phi_q \tag{41}$$

but acquire strings $\phi_p \phi_q$, which are expected to attenuate correlations.

G. Mapping the Hamiltonian

$$c_{p\uparrow}^\dagger c_{q\uparrow} \mapsto a_p^\dagger \phi_{pq} a_q \left(b_p + b_p^\dagger\right)\left(b_q + b_q^\dagger\right)$$

$$4c_{p\downarrow}^\dagger c_{q\downarrow} \mapsto \begin{pmatrix} \left(b_p + b_p^\dagger\right)\left(b_q + b_q^\dagger\right) \\ + \left(b_p + b_p^\dagger\right)\left(b_q - b_q^\dagger\right) e^{\mathrm{i}\pi a_q^\dagger a_q} \\ - \left(b_p - b_p^\dagger\right)\left(b_q + b_q^\dagger\right) e^{\mathrm{i}\pi a_p^\dagger a_p} \\ - \left(b_p - b_p^\dagger\right)\left(b_q - b_q^\dagger\right) e^{\mathrm{i}\pi a_p^\dagger a_p} e^{\mathrm{i}\pi a_q^\dagger a_q} \end{pmatrix}$$

H. Orbital Basis Optimization

The basis can be optimized using techniques similar to those used in oo-pCCD. Guo should broadly explain the MJW orbital optimization here, just to make sure people do not think of it as a bottleneck. Orbital optimization is equivalent to linear canonical transformations of the Majoranas representing the physical fermions.

I. Mean-Field Theory in the Dual Frame

$so(2M)$ HFB. Describe main points only... Cite previous work that is needed to get robust numerics (Wick) etc.

IV. RESULTS AND DISCUSSION

Hubbard plots for energy, overlap with FCI, and expectation values of seniority-zero operators. We want half-filling and doped systems, both repulsive and attractive results.

(Guo: Figures 1-8 are the 6-site Hubbard chain results at half-filling and with hole doping. The MJW-HFB calculations are uLAST-GHFB in the dual representation, where the GHFB and uLAST amplitudes are optimized simultaneously. The MJW-qpCISD calculations are uLAST-qpCISD in the dual representation with fixed mean-field GHFB and simultaneously optimized uLAST amplitudes and CI coefficients.

There are a few issues that I'm still trying to address:

- For the DOCI calculations, I can only get S^2 symmetry-breaking for half-filled repulsive 6-site Hubbard chain. For all the other cases, the UDOCI energies and RDOCI ones are the same. I have tried scanning in different directions, as well as using and not using orbital localization for the initial guesses. I need to explore more initial guesses.

  For attractive systems, my guess is that S^2 does not tend to break since it is energetically favorable for the electrons to pair and localize. However, I don't quite understand why S^2 does not break for DOCI on the 4-electron 6-site repulsive Hubbard chain.

  Tom tried larger hole-doped systems. The S^2 symmetry does break, but the Coulson–Fischer point appears at a large U value.

- For the calculations on the repulsive Hubbard chains, the $N_\uparrow$ and number-parity projected MJW curves are smoother than those without symmetry projection. I've tried using finer grids and starting from multiple random uLAST amplitude guesses. And I'm now running calculations with multiple scans.

  This problem is now fixed. My old Julia run script failed to use the solution of the previous point as the initial guess.

- For attractive systems, the results are much more problematic than those for the repulsive systems. Particularly, the curves of DOCI overlap with FCI are very bumpy. It seems that there could be multiple DOCI solutions that are close in energy but with very different wavefunctions. I'll explore whether we can apply some sort of maximum overlap method.

  The bumpy overlap curves were due to the use of wrong DOCI vector files. This has been fixed for the 6-electron 6-site chain. However, I still see a transition around U = -3. I may need to scan back and forth multiple times when running the DOCI calculations.

)

V. CONCLUSIONS

Previous spin-charge separation mappings did not organize the Clifford algebra according to seniority nor con-

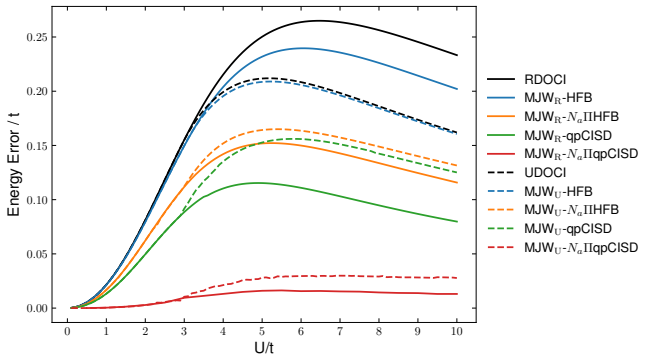

FIG. 1. Energy error for the 6-site Hubbard chain at half-filling. UMJW denotes MJW calculations mapped from UDOCI orbitals.

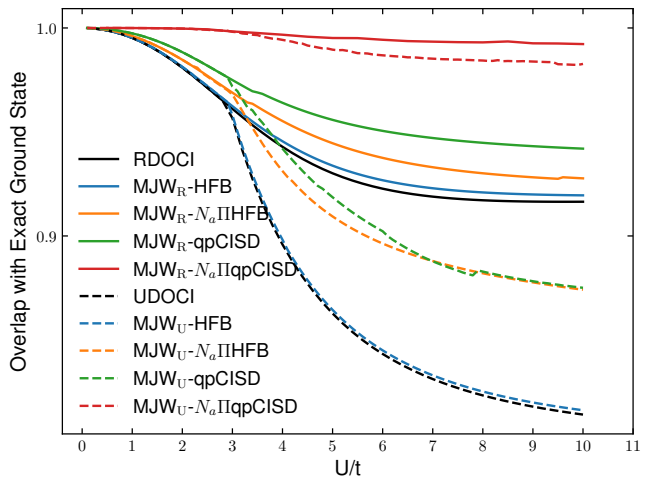

FIG. 2. Overlap with the the FCI wavefunction for the 6-site Hubbard chain at half-filling. UMJW denotes MJW calculations mapped from UDOCI orbitals.

struct a global nonlinear automorphism with selective JW fermionization.

GES: These are bullets in no particular order; many of them obsolete!; some of them may find a better place in R&D

* Explicitly breaking physical Number in the MJW transformation does not mean that physical N will SSB. For example, writing H in a qp basis also breaks N explicitly, yet –per Lieb– there is no SSB.

* But in MJW, Nb breaking, which does not imply physical N broken, means breaking physical seniority, see eqn 28, which of course ought to be beneficial for the energy as we re-entangle a and b fermions.

* Actually, "breaking" physical seniority is not a correct expression; seniority is not a symmetry, so adding nonzero seniority pieces is not symmetry "breaking"; adding nonzero seniority states to the wave function should be beneficial to the energy, but why does it not

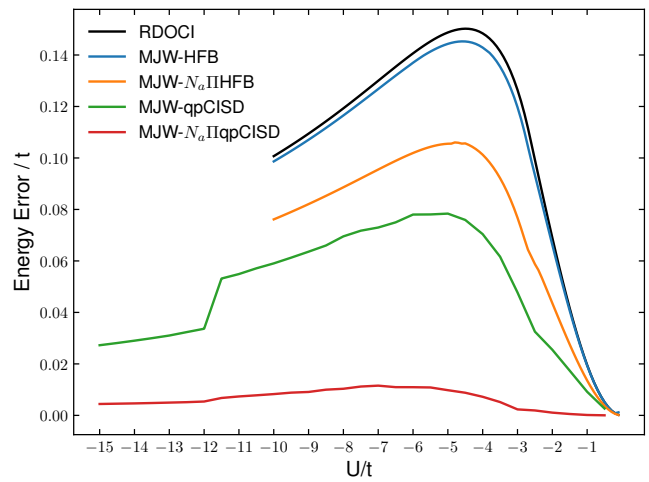

FIG. 3. Energy error for the 6-site Hubbard chain with attractive on-site interaction at half-filling.

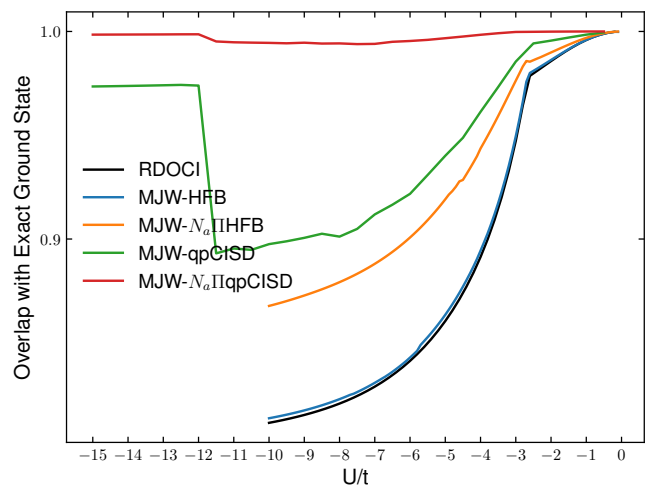

FIG. 4. Overlap with the FCI wavefunction for the 6-site Hubbard chain with attractive on-site interaction at half-filling.

happen with complex qp uLAST in the dual frame ?? (This is now understood from orbital optimization).

* $QRPA$ deserves special mention because we can easily commute the strings through the excitation operators, and is extensive. I think that H^{MJW}_{QRPA} is dense, as opposed to H^{phys}_{QRPA}, where ring and pairing channels do not interact.

* Yet neither $QRPA$ nor $uLAST$ with complex amplitudes has produced the desired result.

* Impact on quantum computation: MJW can be used to map spin 1/2 fermions to qubits by doing the first MJW step and then mapping the spinless fermions to a second $su(2)$; these requires half the number of gates of standard JW because one of the su(2) algebras does not have strings. GES: Maybe this bullet should be expanded

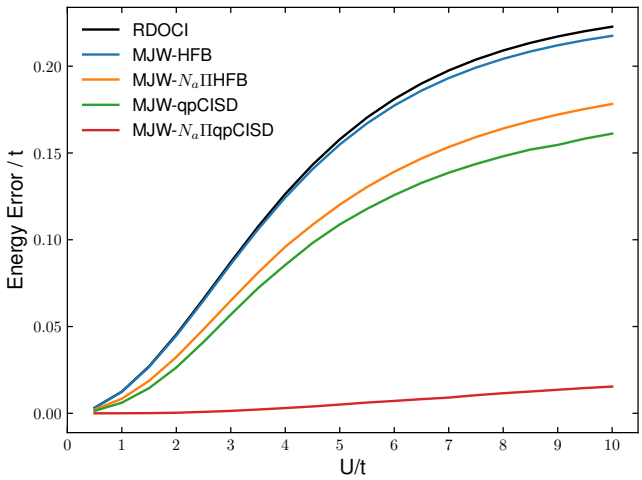

FIG. 5. Energy error for the 4-electron 6-site Hubbard chain.

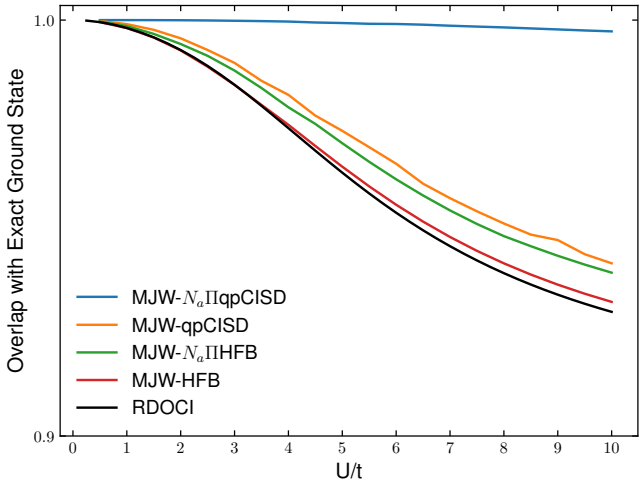

FIG. 6. Overlap with the FCI wavefunction for the 4-electron 6-site Hubbard chain.

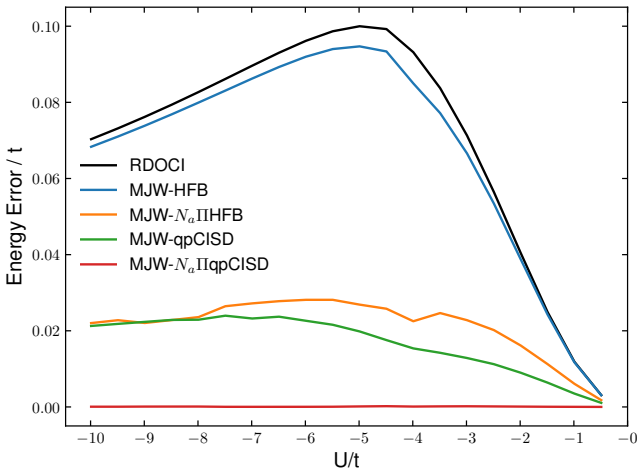

FIG. 7. Energy error for the 4-electron 6-site Hubbard chain with attractive on-site interactions.

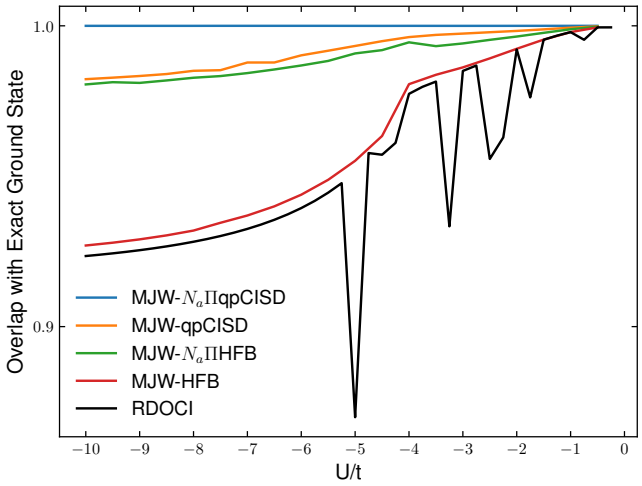

FIG. 8. Overlap with the FCI wavefunction for the 4-electron 6-site Hubbard chain with attractive on-site interactions.

in an Appendix, showing the equations and the strings explicitly

* The mapping is unitary, so the exact ground state's intrinsic bipartite entanglement (for a fixed physical cut) is unchanged by rewriting variables.

* But mean-field is not invariant under such a change of variables: a Gaussian/HFB ansatz in the dual fermions is not the same variational family as a Gaussian/HFB ansatz in the physical fermions, precisely because MJW is nonlinear (Eq. (31))

* Therefore, it can happen that the exact ground state lies "closer" (in overlap) to a dual-Gaussian manifold than to a physical-Gaussian manifold, even though the state itself hasn't changed.

* We thus claim that MJW reorganizes strong correlations into a form more amenable to perturbative treatments.

* The non-Gaussian correlations of the physical problem are, in this dual representation, more efficiently captured by a low-complexity (Gaussian/mean-field + small corrections) ansatz.

* That gives a rigorous "why" behind the heuristic "less entanglement" line: it's really about how much of the state's structure is captured by 2-point data in a chosen operator algebra (Gaussianity relative to that algebra), not about basis-invariant entanglement.

* MJW does not reduce entanglement. It changes its algebraic encoding. That is why mean-field in the dual frame can look "less correlated" even though total entanglement is identical.

* The transformation changes the representation of correlations, not their magnitude.

* Number symmetry breaking allows us to treat strong pairing correlations perturbatively, the same as what we do under spin symmetry breaking. But there is a caveat: N and S^Z become two-body, so projecting them is far from simple. We can project $N_a = \frac{1}{2}N + S^Z$, and physical N and SZ in one shot.

* Introducing attractive terms in model H (like Hirsch 2nd NN density-density attractive interactions) can yield number symmetry breaking, but are these H unitarily related to physical repulsive H? I don't think so. In MJW, the physical number (potentially) breaks spontaneously in the mean-field for an H that has an identical spectrum.

* Under MJW, physical spin symmetry becomes hidden, but other many-body (quasi or dynamical) symmetries may be unraveled. Really?

* Eigenstates of H for finite systems are number-conserving, so if we do a good job at correlating mean-field theory, we should see an improvement in quantum numbers.

* Note Added 14-AUG-25. Most if not all of the above bullets are obsolete after discovering the T1 killer/correlator that creates the nonzero seniority sector of the Hilbert space and affords a clean CC theory to do so, same as in traditional CCD but over a correlated seniority zero state, JW-HF in this case, which is var-pCCD with det coefficients.

Suggested Reviewers:

- Elbio Dagotto
- Gabriel Kotliar
- Joel Moore
- Zohar Nussinov
- Gerardo Ortiz
- Emmanuel Gull
- Steven White

Papers to cite that have used the spin-charge mapping: Chandler and Gunn, arXiv 1997; Brijesh Kumar, PRB 2008; Ostlund and Granath, PRL 96, 066604 (2006); Kurlov et al. in their recent PRB 107, 184202 (2023) use the spin-charge mapping and cite previous works in their Refs 51-56 (I don't think we need to cite those, but we should cite Kurlov).

All of this to make the point that what is novel here is (1) bringing up this well-known mapping in connection with the seniority concept; (2) using JW to build a global, nonlinear canonical transformation which is a bijection, and a unitary automorphism in the full local su(4).

ChatGPT says that the global Parity operator when used to fix a sign is called a Klein in QFT bosonization, and CM folks called it 'Klein factor'; maybe we should call it that too.

ACKNOWLEDGMENTS

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[1] P. Jordan and E. Wigner, Über das Paulische äquivalenzverbot, Zeitschrift für Physik **47**, 631 (1928).

[2] T. M. Henderson, G. P. Chen, and G. E. Scuseria, Strong-weak duality via Jordan-Wigner transformation: using fermionic methods for strongly correlated $su(2)$ spin systems, J. Chem. Phys. **157**, 194114 (2022).

[3] T. M. Henderson, F. Gao, and G. E. Scuseria, Restoring permutational invariance in the Jordan–Wigner transformation, Mol. Phys. **122**, e2254857 (2023).

[4] G. P. Chen and G. E. Scuseria, Robust formulation of Wick's theorem for computing matrix elements between Hartree-Fock-Bogoliubov wavefunctions, J. Chem. Phys. **158**, 231102 (2023).

[5] T. M. Henderson, S. G. Tabrizi, G. P. Chen, and G. E. Scuseria, Hartree-Fock-Bogoliubov theory for number-parity-violating fermionic Hamiltonians, J. Chem. Phys. **160**, 064103 (2024).

[6] C. N. Yang, η pairing and off-diagonal long-range order in a hubbard model, Phys. Rev. Lett. **63**, 2144 (1989).

[7] D. Chandler and J. M. F. Gunn, Mapping spin-charge separation without constraints, Phys. Rev. B **56**, 16430 (1997).

[8] S. Östlund and M. Granath, Exact transformation for spin-charge separation of spin-1/2 fermions without constraints, Phys. Rev. Lett. **96**, 066404 (2006).

[9] B. Kumar, Canonical representation for electrons and its application to the hubbard model, Phys. Rev. B **77**, 205115 (2008).

[10] Y. Wang and T. M. Rice, Exact expression of the t-j model in terms of local spin and fermionic holon operators, Phys. Rev. B **49**, 4360 (1994).

[11] J. Hubbard, Electron correlations in narrow energy bands, Proc. Roy. Soc. London **276**, 238 (1963).

[12] T. L. Allen and H. Shull, Electron pairs in the beryllium atom, J. Phys. Chem. **66**, 2281 (1962).

[13] D. W. Smith and S. J. Fogel, Natural orbitals and geminals of the beryllium atom, J. Chem. Phys. **43**, S91 (1965).

[14] F. Weinhold and E. Bright Wilson Jr., Reduced density matrices of atoms and molecules. i. the 2 matrix of double-occupancy, configuration-interaction wavefunctions for singlet states, J. Chem. Phys. **46**, 2752 (1967).

[15] A. Veillard and E. Clementi, Complete multi-configuration self-consistent field theory, Theoretica Chimica Acta **7**, 133 (1967).

[16] M. Couty and M. B. Hall, Generalized molecular orbital theory ii, J. Phys. Chem. A **101**, 6936 (1997).

[17] C. Kollmar and B. A. Heß, A new approach to density matrix functional theory, J. Chem. Phys. **119**, 4655 (2003).

[18] L. Bytautas, T. M. Henderson, C. A. Jiménez-Hoyos, J. K. Ellis, and G. E. Scuseria, Seniority and orbital symmetry as tools for establishing a full configuration interaction hierarchy, J. Chem. Phys. **135**, 044119 (2011).

[19] T. M. Henderson, I. W. Bulik, T. Stein, and G. E. Scuseria, Seniority-based coupled cluster theory, J. Chem. Phys. **141**, 244104 (2014).